Mats Leander Grobe

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EDUCATION

Technical University of Munich (TUM) — Munich, Germany
M.Sc. in Robotics, Cognition, Intelligence — Oct 2024 – Present
Current Grade Average: 1.4 (Top 2% in TUM School of Computation)

Sungkyunkwan University (SKKU) — Seoul, South Korea
B.Sc. in Mechanical Engineering — Sept 2020 – Aug 2024
Grade Average: 3.99/4.5

Schottengymnasium — Vienna, Austria
Austrian Matura (University Entrance Qualification) — Feb 2015 – July 2019
Grade Average: 1.0 (Summa Cum Laude)

RESEARCH & PROFESSIONAL EXPERIENCE

Practical Course: Deep Learning in Visual Computing — Munich, Germany
TUM Visual Computing & AI Lab (Prof. Matthias Nießner) — Oct 2025 – Present

- Implementing Inverse Rendering pipeline in PyTorch to reconstruct high-fidelity material properties and geometry from OLAT (One-Light-at-a-Time) lightstage data.
- Custom differentiable rasterizer for 2D Gaussian Splatting using CUDA.

Research Project: Neural Rendering (Surface Reconstruction) — Munich, Germany
TUM Visual Computing & AI Lab (Prof. Matthias Nießner) — Apr 2025 – Jul 2025

- Developed "FoamSurf," extending differentiable volumetric rendering framework RadFoam by integrating monocular depth/normal priors into CUDA raytracing pipeline.
- Outperformed 2D Gaussian Splatting in mesh reconstruction quality (MAE) for surface extraction tasks. **[Click here for Code.]**

Tutor: Introduction to Deep Learning (I2DL) — Munich, Germany
Technical University of Munich — Apr 2025 – Present

- Teaching Deep Learning concepts (Backpropagation, CNNs, Transformers, GANs) to graduate students.
- Assisting with PyTorch implementation exercises and grading exams.

Researcher: Advanced Additive Manufacturing Lab — Seoul, South Korea
Sungkyunkwan University (Prof. Brian J. Lee) — Nov 2023 – Mar 2024

- Full instrumentation of high-precision CLIP (Continuous Liquid Interface Production) 3D Printer.
- Developed the control software in C++ from the ground up, including a custom GUI via Qt.
- Achieved 20-micron print resolution through precise stepper motor control. **[Click here for Code.]**

Intern: Embedded Systems — Seoul, South Korea
MAN Truck & Bus Korea — June 2023 – July 2023

- Designed a Diesel Engine Control Unit (ECU) interface using Arduino microcontrollers.
- Enabled two-way communication via CAN Bus to control truck features for diagnostics.
- Authored technical manuals for local engineering staff.

Researcher: Microfluidic Convergence Lab — Seoul, South Korea
Sungkyunkwan University (Prof. Jinkee Lee) — Dec 2022 – Feb 2023

- Conducted Computational Fluid Dynamics (CFD) simulations of gas-dynamic virtual nozzles.
- Analyzed droplet formation behavior using simulation data to optimize nozzle geometry.

Intern: Robotics Support Seoul, South Korea
Plasmatreat Korea Dec 2021 – Feb 2022

- Programmed a 3-axis robot for automated atmospheric plasma surface treatment in a lab environment.
- Supported the sales engineering team during on-site customer visits and ad-hoc service calls.

SKILLS & LANGUAGES

Programming & Compute: C++ (STL, C++17), Python, CUDA, PyTorch, MATLAB, Qt.
Simulation & Engineering: CFD (Ansys Fluent), ROS/ROS2, Arduino, CAN Bus, Git, Docker, Linux.
Languages:

- **German:** Native
- **English:** Expert - C1/C2 (TOEFL 112, IELTS 8.0)
- **Korean:** University Entrance Qualification - B2 (TOPIK 4)
- **Russian:** Intermediate
- **Italian:** Intermediate

VOLUNTEERING

European Solidarity Corps Samara, Russia
Volunteer in Kindergarten (Makkabi Lauder) Sept 2019 – Mar 2020